

LAZARD LEGACY MATH ACADEMY

Geometry Basics Practice

Quick Drill — 30 Minutes

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"Dreams can come true. Better late than never."

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TABLE OF CONTENTS

1. Quick Lesson: Angles (Complementary, Supplementary, Vertical)
2. Quick Lesson: Area & Perimeter (Rectangle, Triangle, Circle)
3. Quick Lesson: Pythagorean Theorem
4. Practice Drill: 40 Mixed Geometry Problems
5. Answer Key
6. Student Notes

Section 1: Quick Lesson — Angles

Key Definitions:

- **Complementary Angles:** Two angles whose measures add up to exactly 90° .
Example: 30° and 60° are complementary.
- **Supplementary Angles:** Two angles whose measures add up to exactly 180° .
Example: 120° and 60° are supplementary.
- **Vertical Angles:** When two lines cross, the angles across from each other are equal. They are also called 'opposite angles.'
- **Adjacent Angles:** Angles that share a common side and vertex.

Quick Memory Tips:

- Complementary → think 'Corner' (90° like a corner)
- Supplementary → think 'Straight line' (180°)
- Vertical → think 'V for Victory' (the equal pair wins)

Worked Examples:

Example 1:

Two angles are complementary. One measures 35° . Find the other.

Solution: Complementary angles sum to 90° . So: $90^\circ - 35^\circ = 55^\circ$. The other angle is 55° .

Example 2:

Two angles are supplementary. One measures 110° . Find the other.

Solution: Supplementary angles sum to 180° . So: $180^\circ - 110^\circ = 70^\circ$. The other angle is **70°** .

Example 3:

Two vertical angles are formed by intersecting lines. One is 48° . What is the other?

Solution: Vertical angles are always equal. So the other angle is also **48°** . (The adjacent angles would be $180^\circ - 48^\circ = 132^\circ$.)

Example 4:

Angle A and Angle B are complementary. Angle A = $2x$ and Angle B = $x + 15$. Find x and both angles.

Solution: Set up: $2x + (x + 15) = 90$. Combine: $3x + 15 = 90$. Subtract 15: $3x = 75$. Divide by 3: $x = 25$. So Angle A = $2(25) = \mathbf{50^\circ}$ and Angle B = $25 + 15 = \mathbf{40^\circ}$. Check: $50 + 40 = 90$ ✓

Example 5:

Two supplementary angles have measures $(3x + 10)^\circ$ and $(2x - 5)^\circ$. Find both angles.

Solution: Set up: $(3x + 10) + (2x - 5) = 180$. Combine: $5x + 5 = 180$. Subtract 5: $5x = 175$. Divide by 5: $x = 35$. First angle = $3(35) + 10 = \mathbf{115^\circ}$. Second angle = $2(35) - 5 = \mathbf{65^\circ}$. Check: $115 + 65 = 180$ ✓

Section 2: Quick Lesson — Area & Perimeter

Essential Formulas:

Rectangle:

- Area = length $\times$ width ($A = l \times w$)
- Perimeter = 2(length) + 2(width) ($P = 2l + 2w$)

Triangle:

- Area = $\frac{1}{2} \times$ base $\times$ height ($A = \frac{1}{2}bh$)
- Perimeter = sum of all three sides ($P = a + b + c$)

Circle:

- Area = $\pi \times$ radius² ($A = \pi r^2$)
- Circumference = $2 \times \pi \times$ radius ($C = 2\pi r$)
- Remember: diameter = $2 \times$ radius, so $r = d/2$

Worked Examples:

Example 1:

Find the area and perimeter of a rectangle with length 8 cm and width 5 cm.

Solution: Area = $l \times w = 8 \times 5 = \mathbf{40 \text{ cm}^2}$. Perimeter = $2(8) + 2(5) = 16 + 10 = \mathbf{26 \text{ cm}}$.

Example 2:

Find the area of a triangle with base 12 in and height 7 in.

Solution: Area = $\frac{1}{2} \times$ base $\times$ height = $\frac{1}{2} \times 12 \times 7 = \frac{1}{2} \times 84 = \mathbf{42 \text{ in}^2}$.

Example 3:

Find the area and circumference of a circle with radius 6 m. (Use $\pi \approx 3.14$)

Solution: Area = $\pi r^2 = 3.14 \times (6)^2 = 3.14 \times 36 = \mathbf{113.04 \text{ m}^2}$. Circumference = $2\pi r = 2 \times 3.14 \times 6 = \mathbf{37.68 \text{ m}}$.

Example 4:

A rectangular garden is 15 ft by 9 ft. How much fencing is needed to surround it?

Solution: We need the perimeter. $P = 2(15) + 2(9) = 30 + 18 = \mathbf{48 \text{ ft}}$ of fencing needed.

Example 5:

Find the area of a triangle with base 10 m and height 4.5 m.

Solution: Area = $\frac{1}{2} \times 10 \times 4.5 = \frac{1}{2} \times 45 = \mathbf{22.5 \text{ m}^2}$.

Section 3: Quick Lesson — Pythagorean Theorem

The Pythagorean Theorem: $a^2 + b^2 = c^2$

- This works ONLY for right triangles (triangles with a 90° angle).
- 'a' and 'b' are the two legs (shorter sides).
- 'c' is the hypotenuse (the longest side, across from the right angle).

Common Pythagorean Triples (memorize these!):

- 3, 4, 5 (and multiples: 6-8-10, 9-12-15, 12-16-20)
- 5, 12, 13 (and multiples: 10-24-26)
- 7, 24, 25
- 8, 15, 17

Worked Examples:

Example 1:

A right triangle has legs of 3 and 4. Find the hypotenuse.

Solution: $c^2 = a^2 + b^2 = 3^2 + 4^2 = 9 + 16 = 25$. Take the square root: $c = \sqrt{25} = 5$.

Example 2:

A right triangle has legs of 5 and 12. Find the hypotenuse.

Solution: $c^2 = 5^2 + 12^2 = 25 + 144 = 169$. $c = \sqrt{169} = 13$.

Example 3:

The hypotenuse is 10 and one leg is 6. Find the other leg.

Solution: $a^2 + 6^2 = 10^2$. $a^2 + 36 = 100$. $a^2 = 100 - 36 = 64$. $a = \sqrt{64} = 8$.

Example 4:

A ladder leans against a wall. The base is 8 ft from the wall, and the ladder is 17 ft long. How high does it reach?

Solution: The ladder is the hypotenuse. $h^2 + 8^2 = 17^2$. $h^2 + 64 = 289$. $h^2 = 225$. $h = \sqrt{225} = \mathbf{15 \text{ ft}}$.

Example 5:

A right triangle has legs of 7 and 24. Find the hypotenuse.

Solution: $c^2 = 7^2 + 24^2 = 49 + 576 = 625$. $c = \sqrt{625} = \mathbf{25}$.

Section 4: Practice Drill — 40 Mixed Geometry Problems

Directions: Solve each problem. Show your work in the space provided. Use $\pi \approx 3.14$ where needed.

Angles (Problems 1–10):

1. Two complementary angles: one is 22° . Find the other.
2. Two supplementary angles: one is 135° . Find the other.
3. Vertical angles are formed. One is 67° . Find the other.
4. Angles A and B are complementary. $A = 4x$, $B = x + 10$. Find both angles.
5. Angles C and D are supplementary. $C = 2x + 30$, $D = x$. Find both angles.
6. Two supplementary angles: one is 3 times the other. Find both.
7. Two complementary angles: one is 5 times the other. Find both.
8. Two vertical angles are $(2x + 10)^\circ$ and $(3x - 15)^\circ$. Find x and the angle measures.
9. Two supplementary angles: one is 20° more than the other. Find both.

10. Two complementary angles: $(x + 5)^\circ$ and $(2x - 8)^\circ$. Find both angles.

Area & Perimeter (Problems 11–25):

- 11.** Find the area of a rectangle: $l = 14$ cm, $w = 6$ cm.
- 12.** Find the perimeter of a rectangle: $l = 9$ m, $w = 4$ m.
- 13.** Find the area of a triangle: $b = 8$ in, $h = 5$ in.
- 14.** Find the circumference of a circle: $r = 10$ ft.
- 15.** Find the area of a circle: $r = 3$ cm.
- 16.** Find the perimeter of a triangle with sides 5, 12, and 13.
- 17.** Find the area of a rectangle: $l = 20$ ft, $w = 11$ ft.
- 18.** Find the area of a circle with diameter 14 m.
- 19.** A square has side 9 cm. Find its area and perimeter.
- 20.** A rectangle has area 72 cm^2 . Width = 8 cm. Find the length.
- 21.** Find the perimeter of a rectangle with $l = 25$ m, $w = 12$ m.
- 22.** Find the area of a triangle: $b = 16$ ft, $h = 9$ ft.

23. Find the circumference of a circle with $r = 7$ in.

24. A rectangle has perimeter 50 cm. Length is 15 cm. Find width.

25. Find the area and perimeter of a rectangle: $l = 12$ in, $w = 8$ in.

Pythagorean Theorem (Problems 26–40):

- 26.** A right triangle has legs 6 and 8. Find the hypotenuse.
- 27.** A right triangle has legs 9 and 12. Find the hypotenuse.
- 28.** Hypotenuse = 13, one leg = 5. Find the other leg.
- 29.** Hypotenuse = 26, one leg = 10. Find the other leg.
- 30.** A right triangle has legs 8 and 15. Find the hypotenuse.
- 31.** Find the hypotenuse: legs = 20 and 21.
- 32.** A right triangle has hypotenuse 15 and leg 9. Find the other leg.
- 33.** A triangle has sides 3, 4, and 5. Is it a right triangle? Explain.
- 34.** Find the hypotenuse: legs = 11 and 60.
- 35.** A right triangle has legs 10 and 24. Find the hypotenuse.
- 36.** Find the area of a triangle: $b = 20$ m, $h = 13$ m.
- 37.** Find the circumference of a circle with diameter 20 cm.

38. Two supplementary angles have ratio 2:3. Find both.

39. A right triangle has hypotenuse 25 and one leg 7. Find the other leg.

40. A ladder is 20 ft long. Base is 12 ft from wall. How high does it reach?

Section 5: Answer Key

1. 68°
2. 45°
3. 67°
4. $x = 16$; $A = 64^\circ$, $B = 26^\circ$
5. $x = 50$; $C = 130^\circ$, $D = 50^\circ$
6. 45° and 135°
7. 15° and 75°
8. $x = 25$; each angle = 60°
9. 80° and 100°
10. $x = 31$; angles = 36° and 54°
11. 84 cm^2
12. 26 m
13. 20 in^2
14. 62.8 ft
15. 28.26 cm^2
16. 30
17. 220 ft^2
18. 153.86 m^2
19. Area = 81 cm^2 , Perimeter = 36 cm
20. 9 cm
21. 74 m
22. 72 ft^2
23. 43.96 in
24. 10 cm
25. Area = 96 in^2 , Perimeter = 40 in

26. 10

27. 15

28. 12

29. 24

30. 17

31. 29

32. 12

33. Yes: $3^2 + 4^2 = 9 + 16 = 25 = 5^2$

34. 61

35. 26

36. 130 m^2

37. 62.8 cm

38. 72° and 108°

39. 24

40. 16 ft

